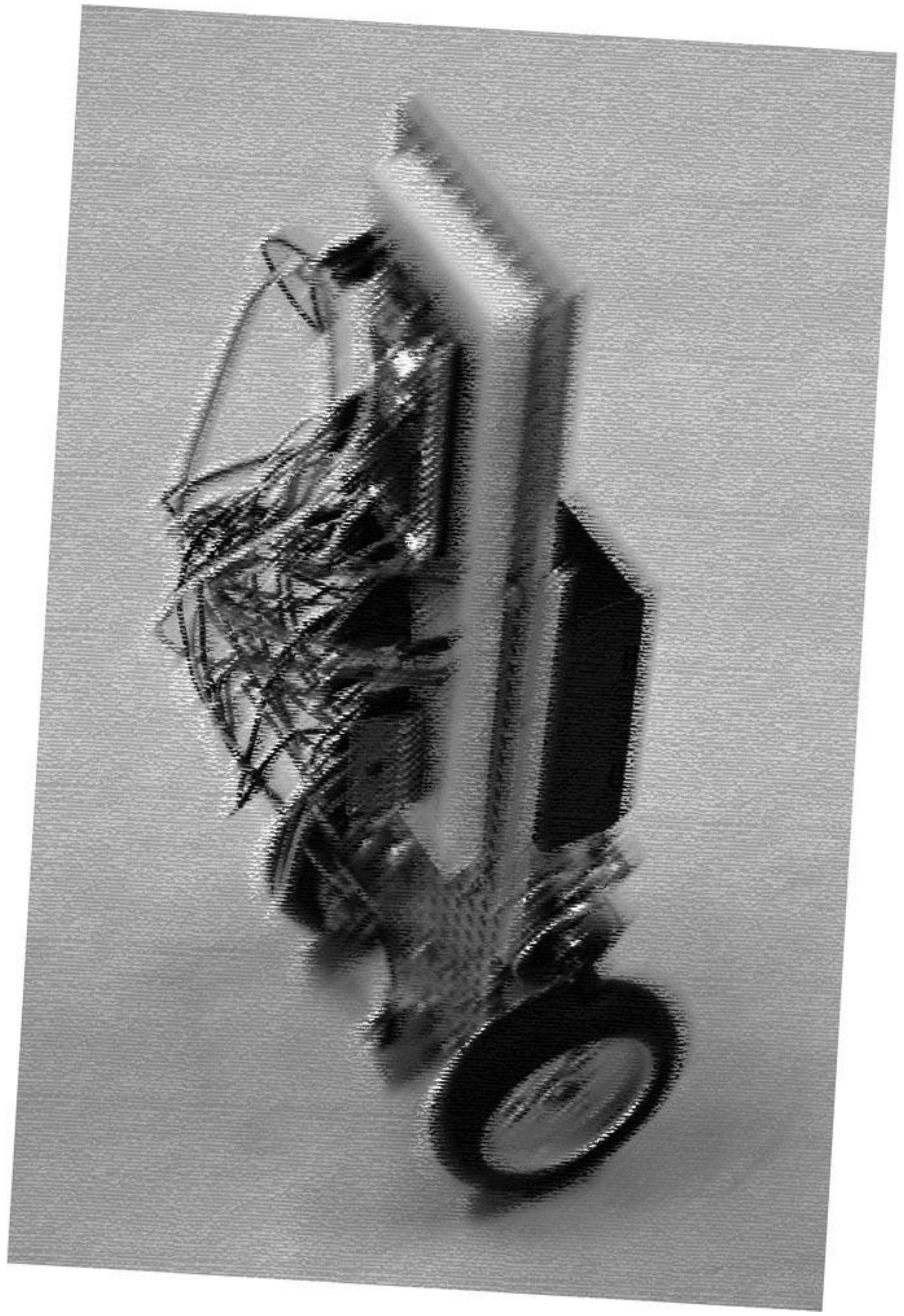
PRACTICAL EXERCISE · WEEK 1

Construction of a Wheeled Inverted Pendulum

Shaoqi (シャオ チー)

COURSE	Department of Precision Engineering · 2026 S1S2
SESSIONS	Mondays, 13:00–18:00 · Room 326, Engineering Bldg. 14
TODAY	Course Introduction — 10 minutes

V1.2 2026·04·27



Week 0 - You Should Have Done These.

- ✓ Register a Keil Studio Cloud Account
- ✓ Install Processing 4
- ✓ Install Fritzing
- ✓ Download the Course Material

No worries if you don’t get it done.

THE FOUR-WEEK ARC

Timeline

1

Week 1
Standing

Assembly & bring-up

Angle-only **standing** demo

2

Week 2
Position control

Build a rotary encoder

Position control

Simulation in Simulink

3

Week 3
EXTENSION

Load & disturbance tests

Advanced control (LQR/MPC)

……

4

Week 4
presentation

10-minute presentation

Live demo & Q&A

Final grade · 40%

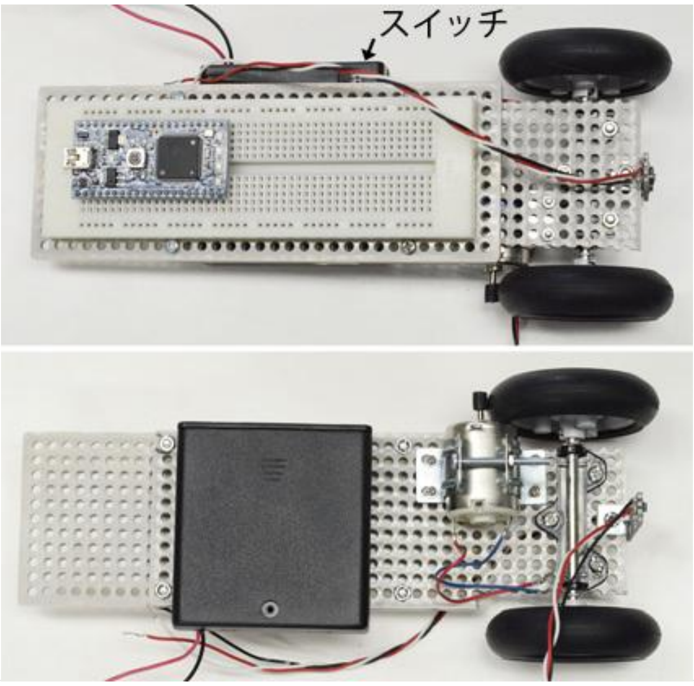
Step 0 - Work in parallel.

Assign the tasks among yourselves. If there are four people in the group, two people can work on one task together. Alternatively, the fourth member can begin circuit assembly.

A TASK A

STUDENT A · MECHANICAL

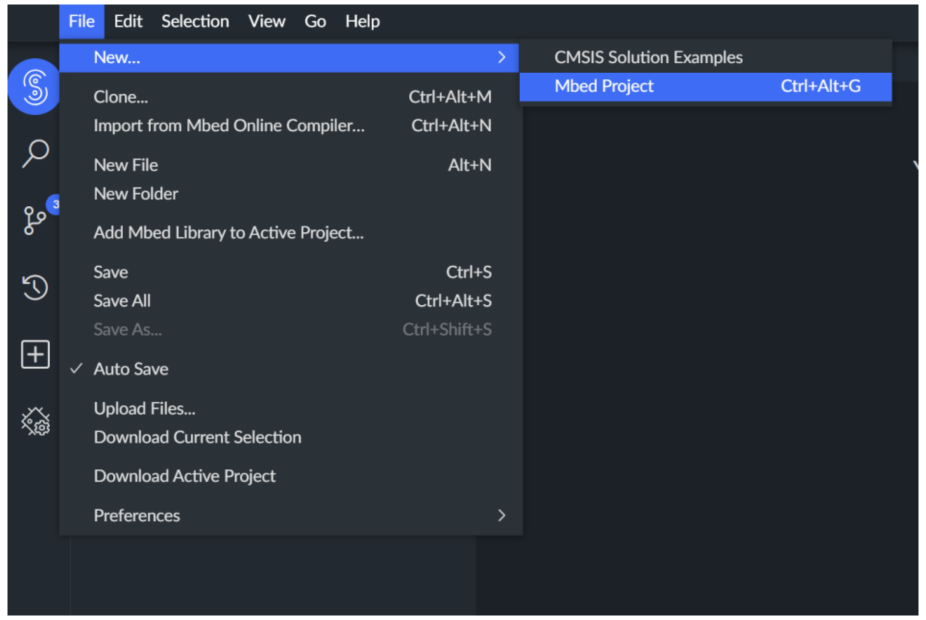
Mechanical assembly



B TASK B

STUDENT B · FIRMWARE

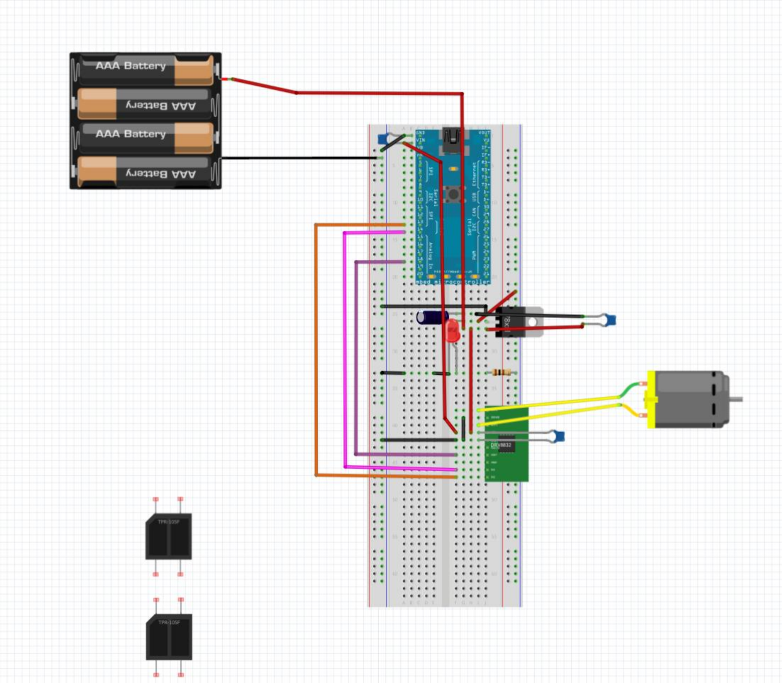
Microcontroller



C TASK C

STUDENT C · WIRING DESIGN

Fritzing breadboard layout



Microcontroller

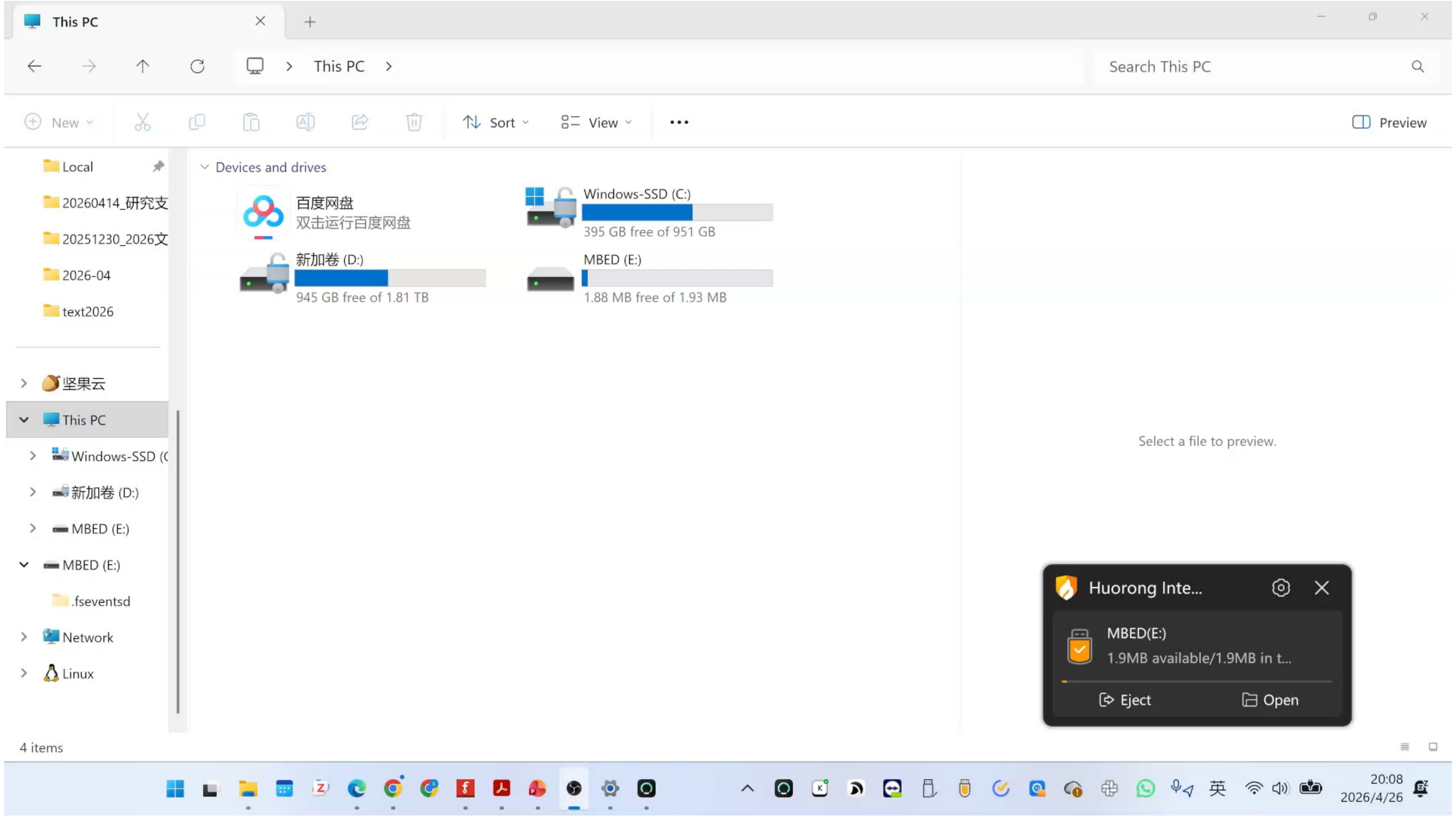
course_materials\debug_tools

D1_motor_controller.bin
compiled binary
generated by Keil Cloud → Copy to Microcontroller

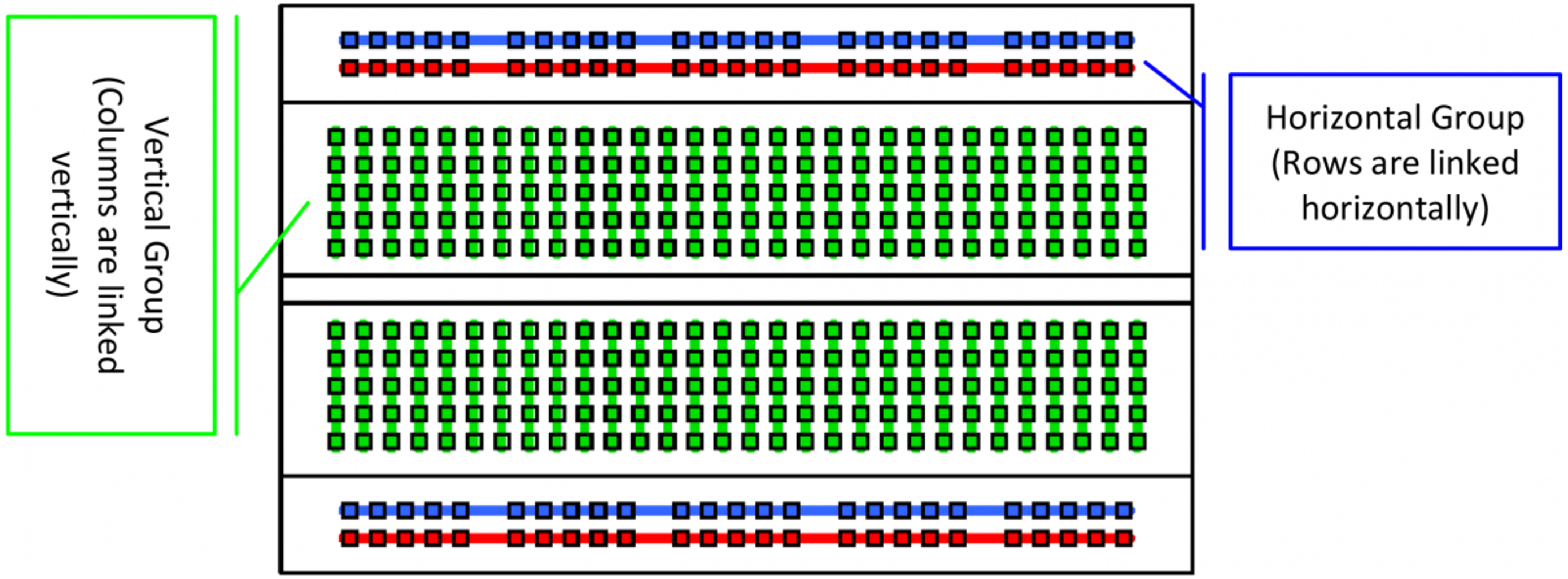
D1_motor_controller.cpp
source code
Open with Notepad → Copy to Keil Cloud

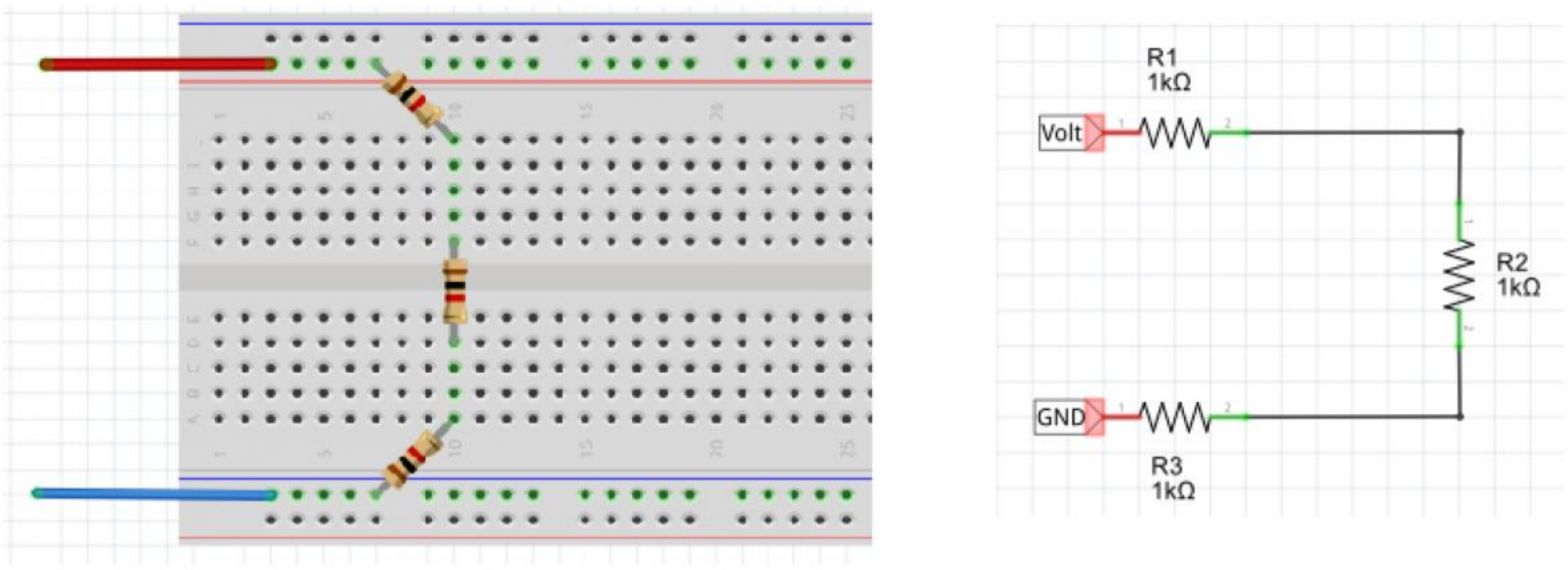
D1_motor_controller.pde
debugging tool

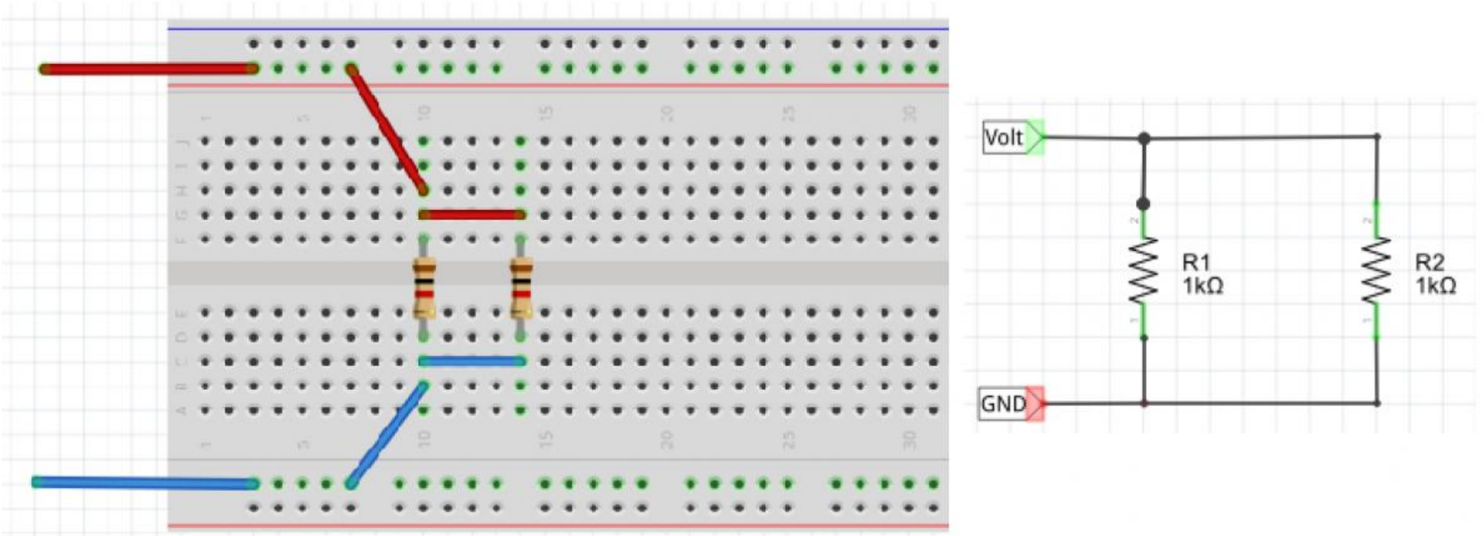
Microcontroller



Breadboard

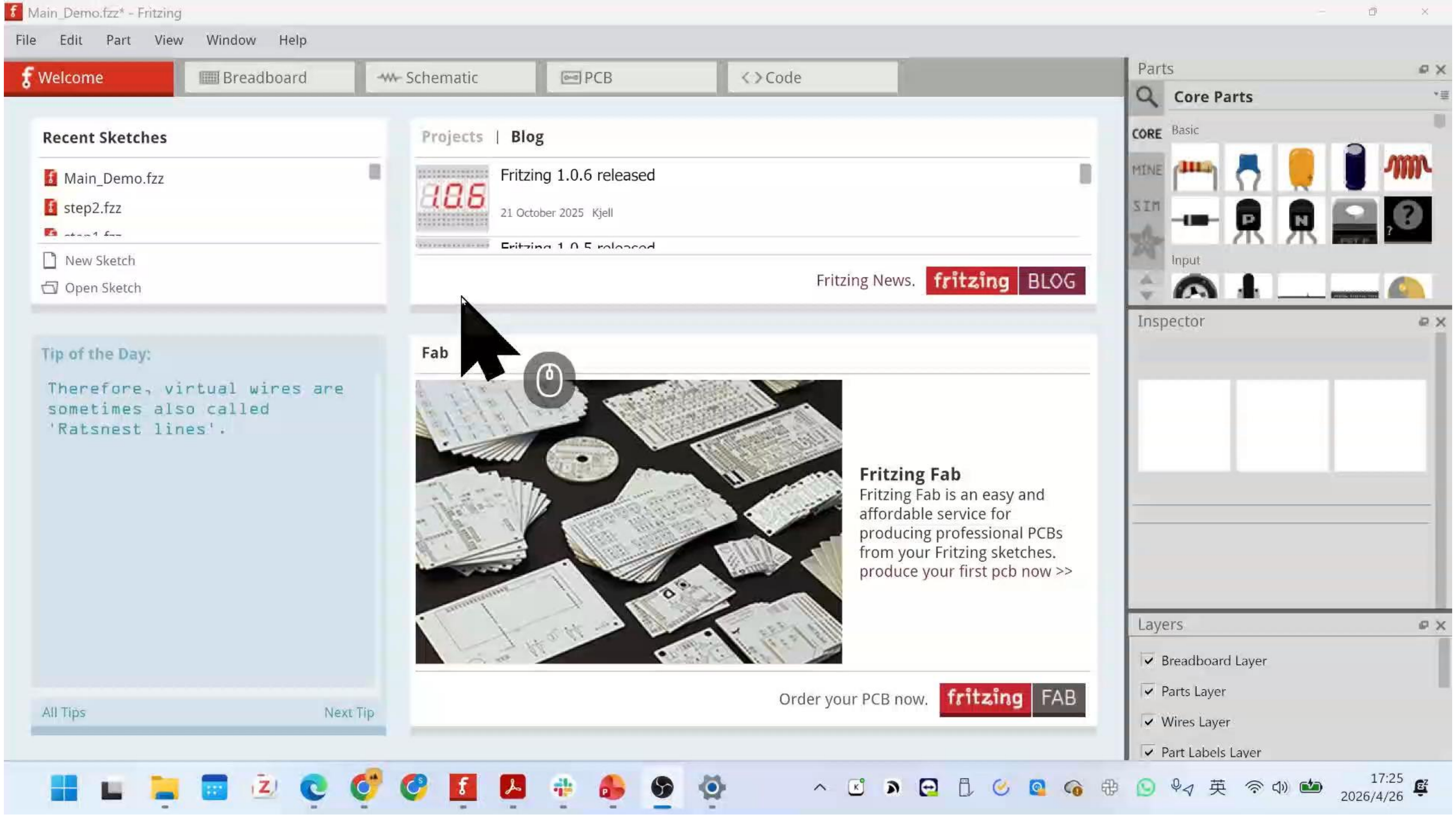






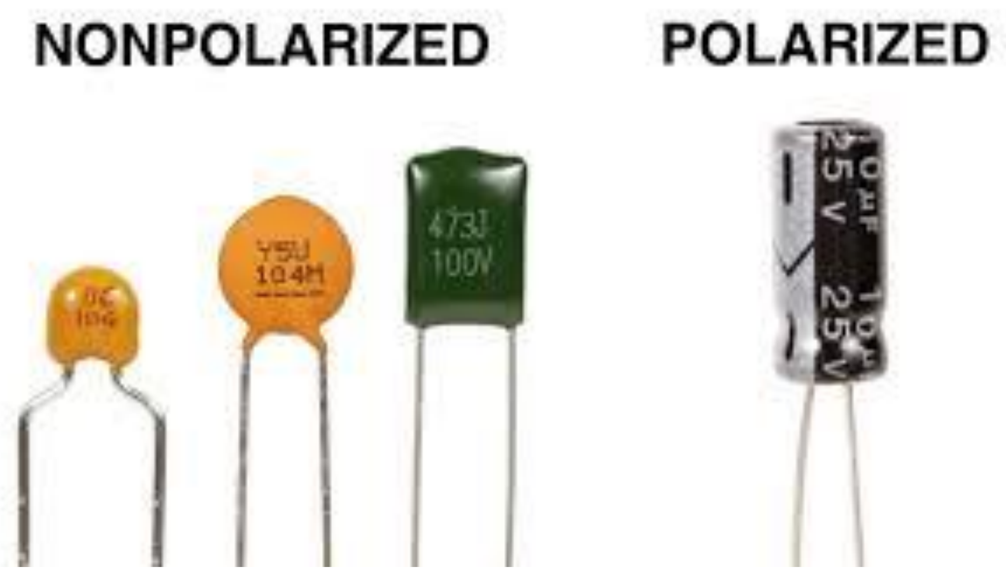
It’s quite abstract.

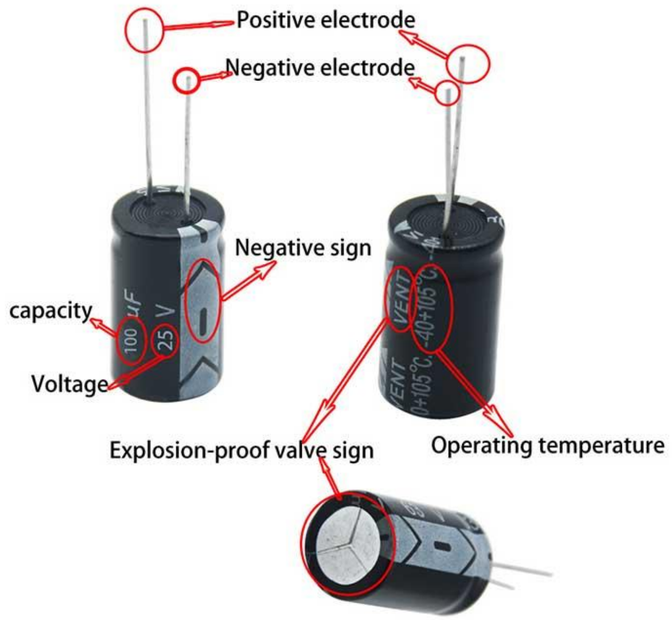
Fritzing-assisted breadboard layout

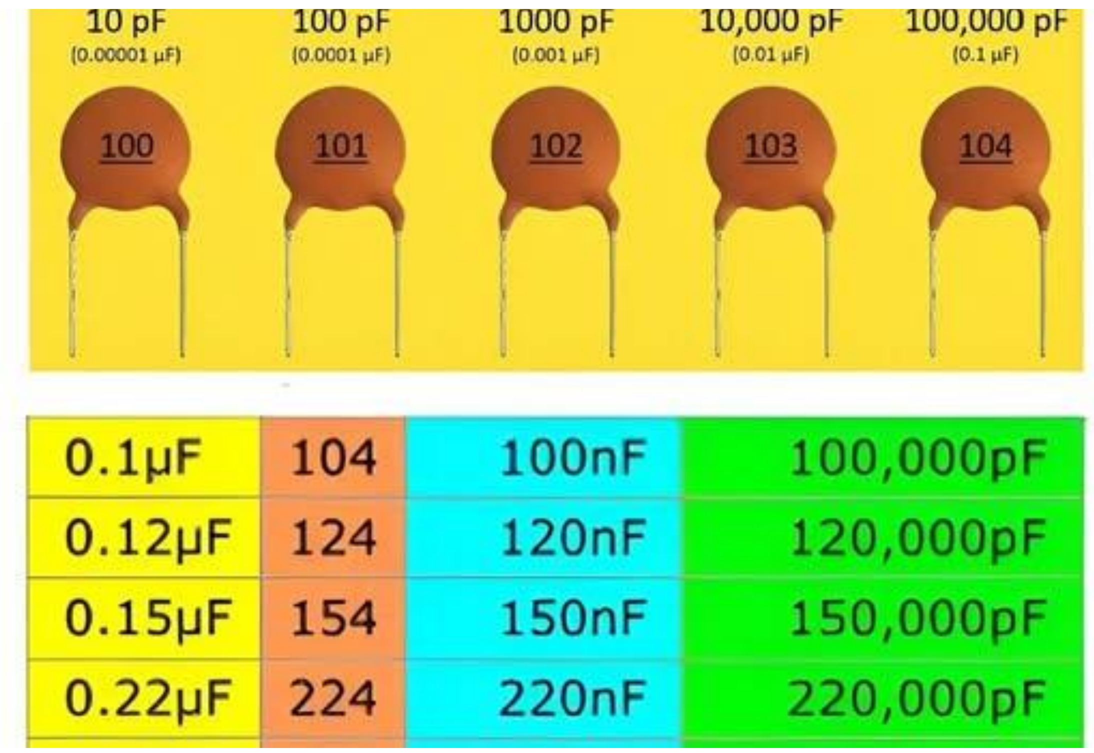


Component values and polarity

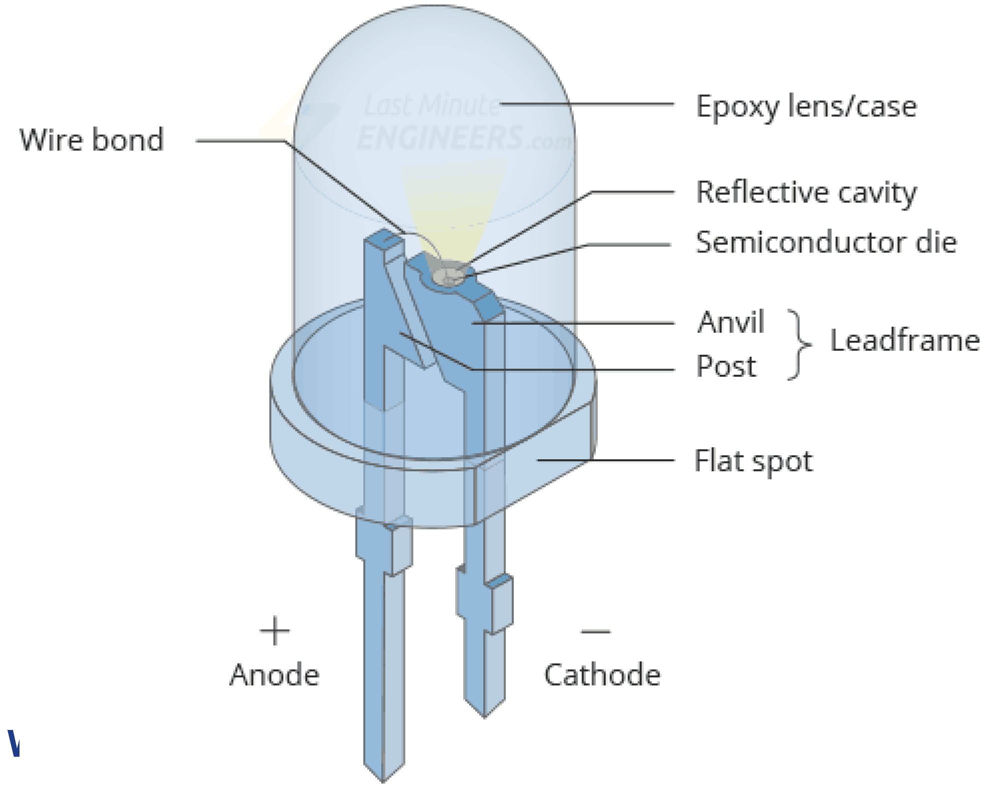
$104 = 10 \times 10^4 = 100{,}000\ \text{pF} = 0.1\ \mu\text{F}$

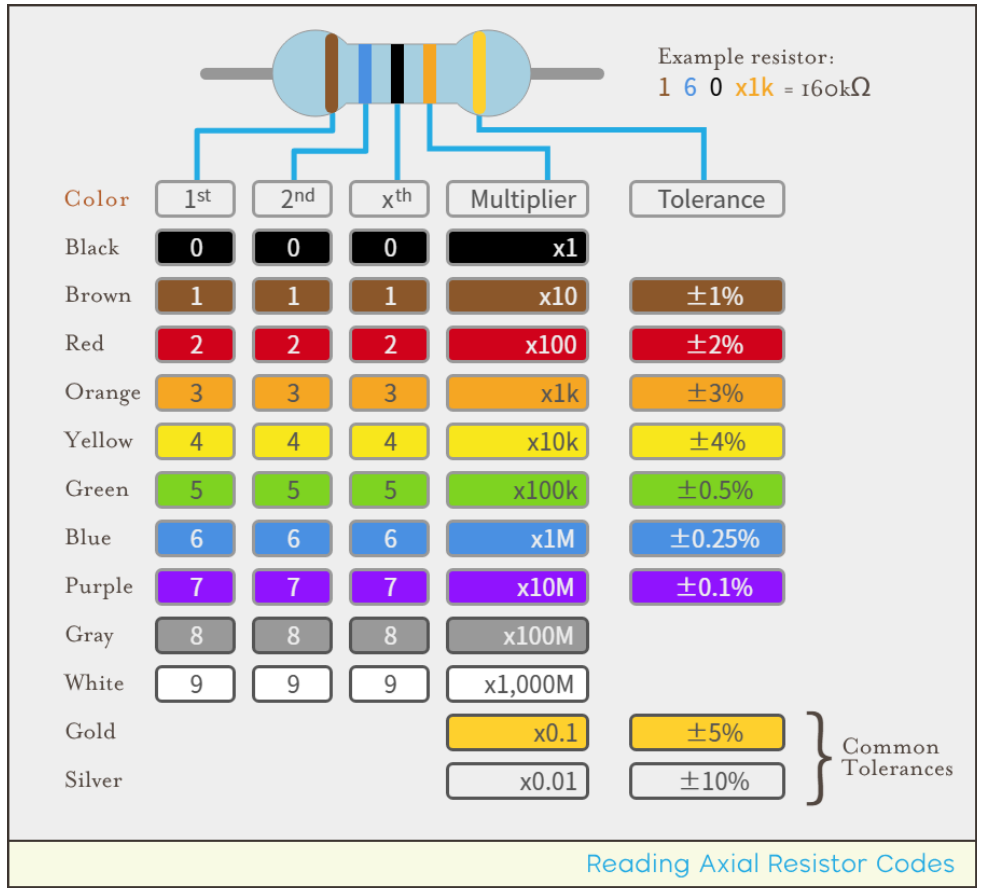






0.1µF	104	100nF	100,000pF
0.12µF	124	120nF	120,000pF
0.15µF	154	150nF	150,000pF
0.22µF	224	220nF	220,000pF

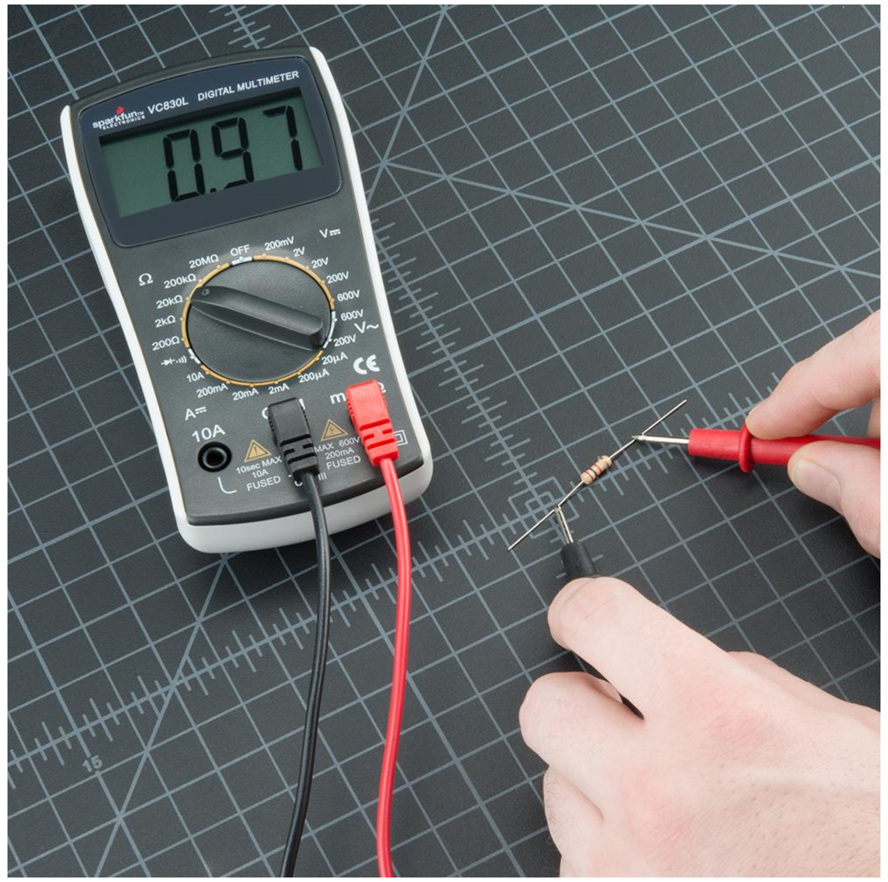




Color	1st	2nd	xth	Multiplier	Tolerance
Black	0	0	0	x1	
Brown	1	1	1	x10	±1%
Red	2	2	2	x100	±2%
Orange	3	3	3	x1k	±3%
Yellow	4	4	4	x10k	±4%
Green	5	5	5	x100k	±0.5%
Blue	6	6	6	x1M	±0.25%
Purple	7	7	7	x10M	±0.1%
Gray	8	8	8	x100M	
White	9	9	9	x1,000M	
Gold				x0.1	±5%
Silver				x0.01	±10%

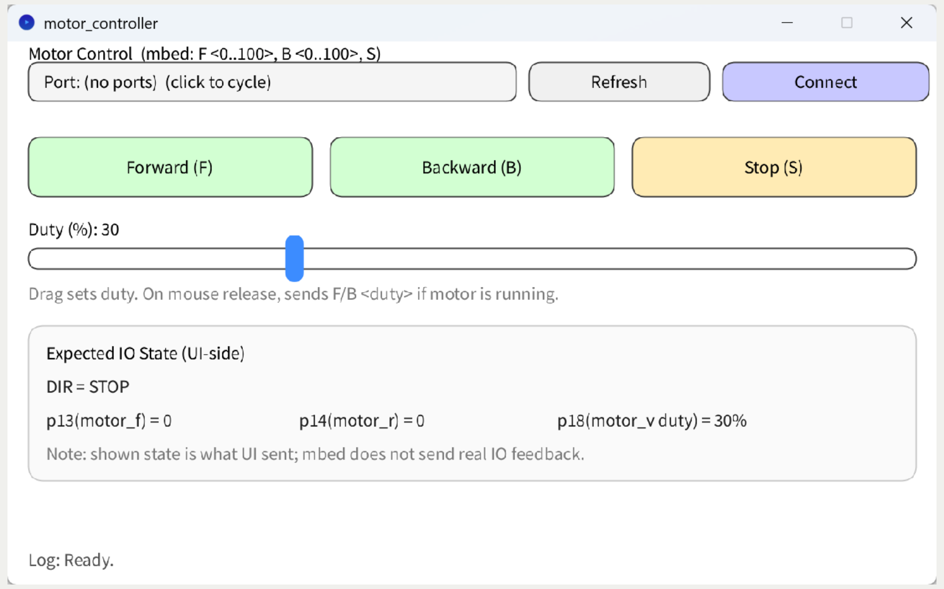
Common Tolerances (Gold, Silver)

Reading Axial Resistor Codes

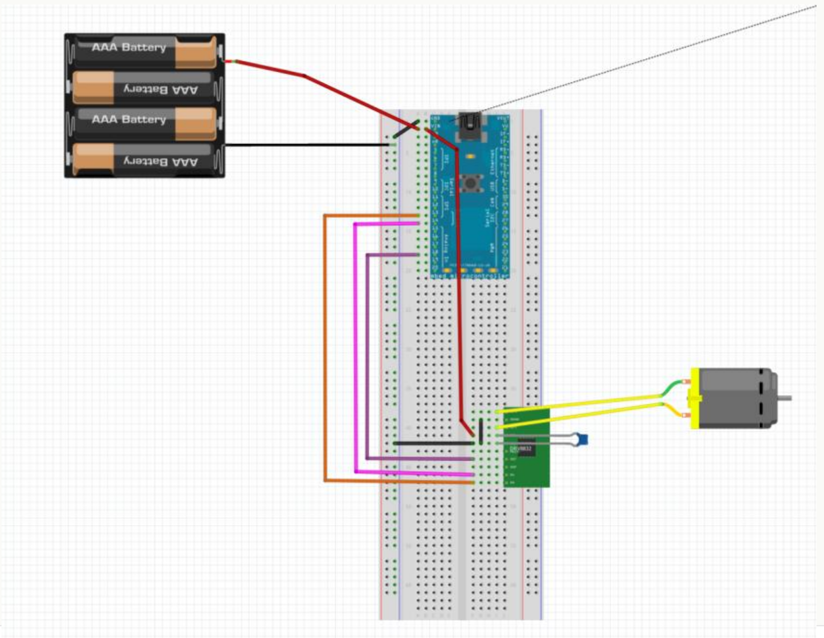


Team Collaboration (3 STEP)

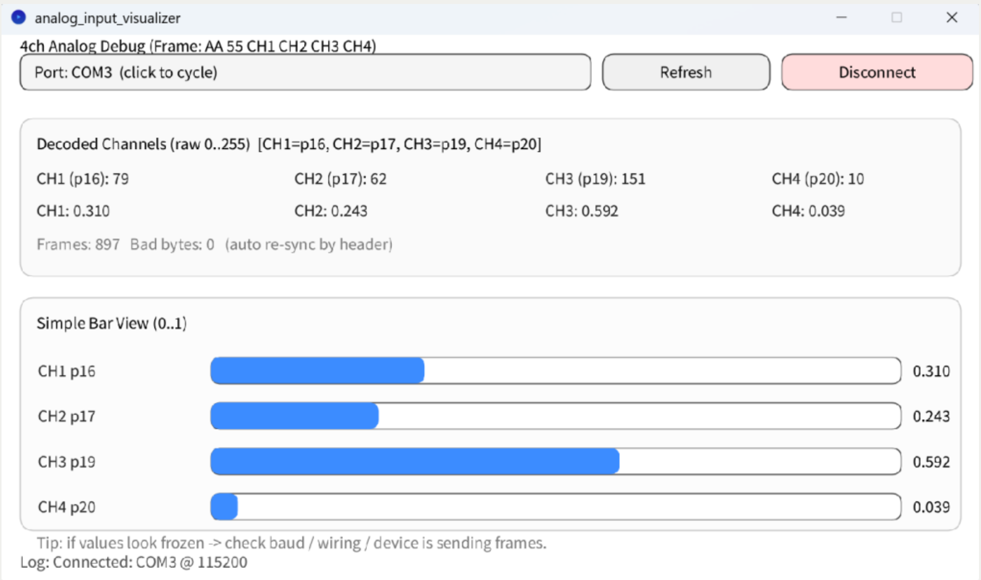
TOOL 115200



STEP 1
Motor Controller

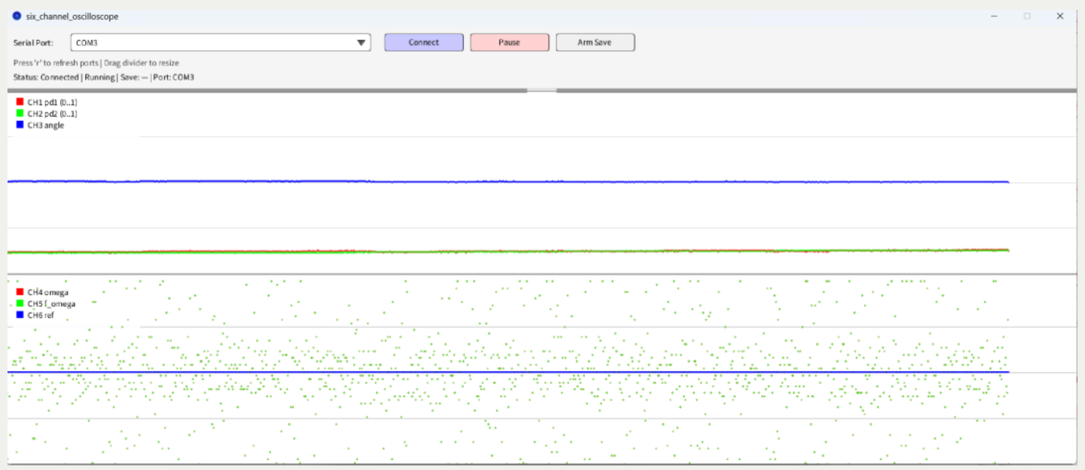


TOOL 115200



STEP 2
Analog Input Visualizer

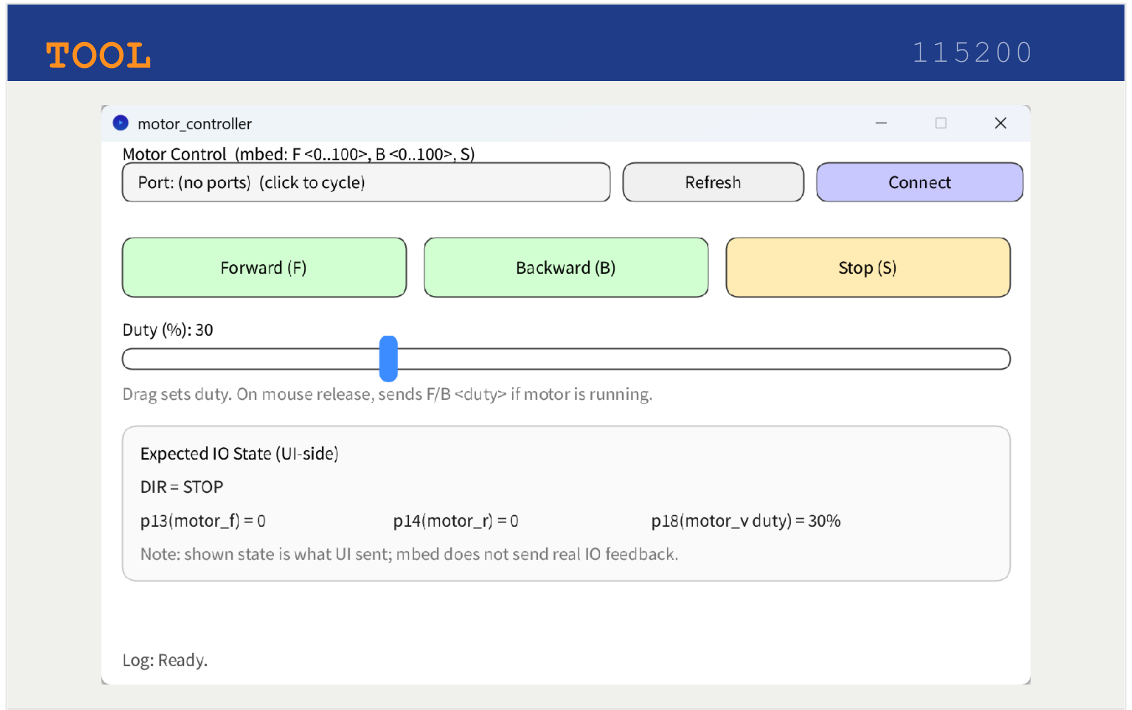
TOOL 230400



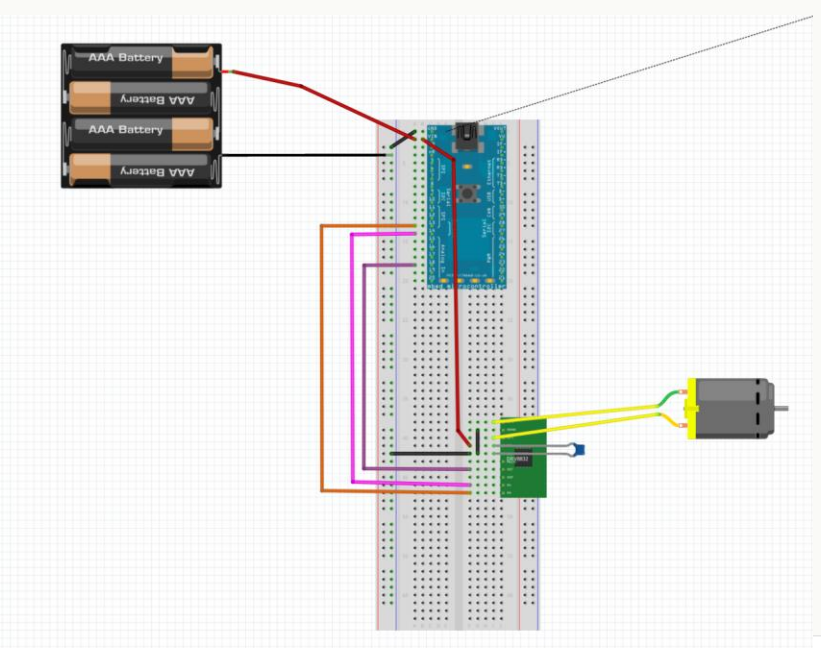
STEP 3 ·
Standing Example

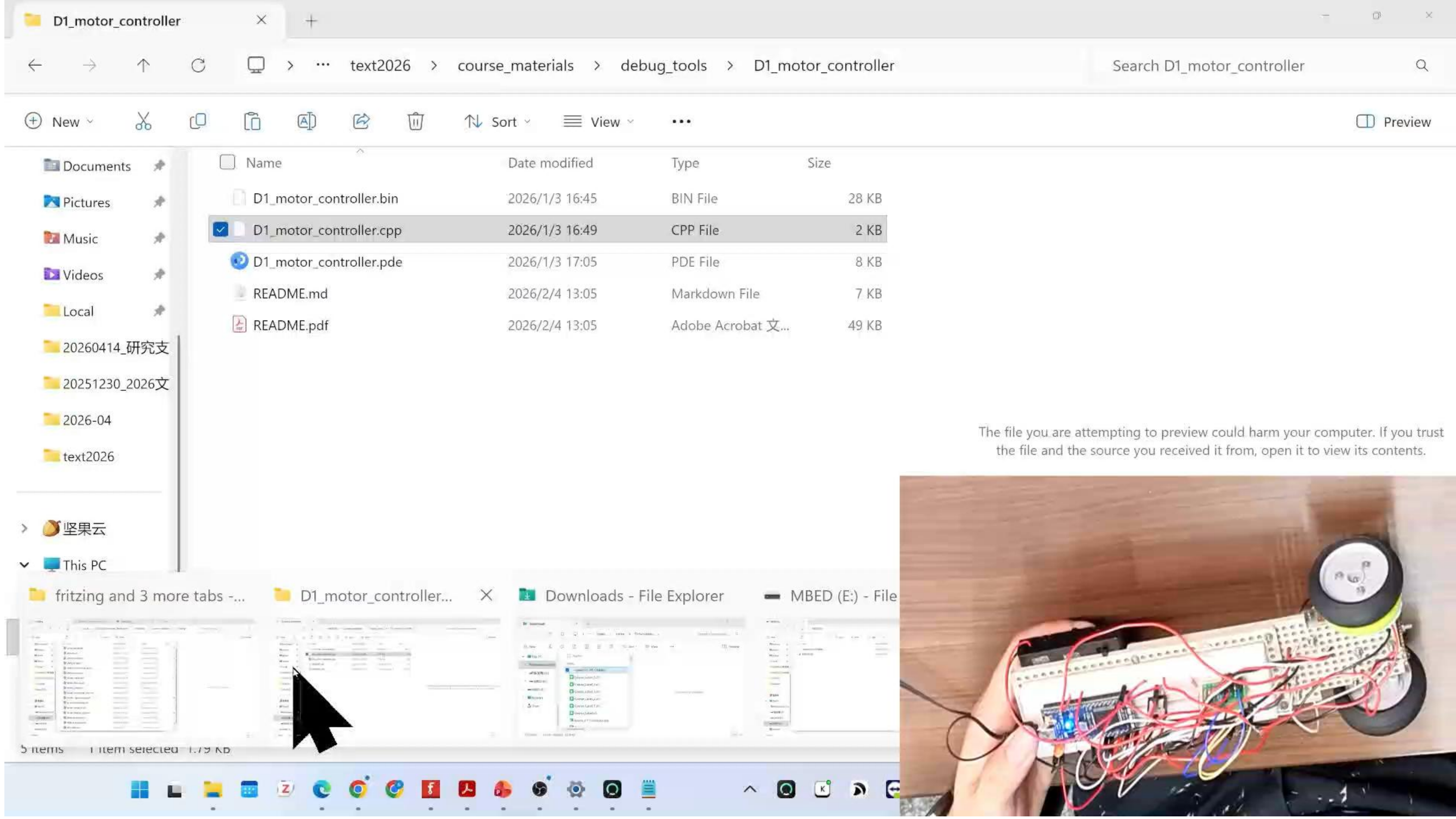
AI tools may be used for editing and refinement.

Step 1 – Motor Controller



STEP 1
Motor Controller





Grading Criteria

Weekly submissions · 60% | **Final Presentation· 40%**

Week	Deliverables	Score (%)	Notes
W1	— One video: standing demo	**60%** (Weeks 1–3 submissions)	Due by **9 pm** on the day of the session. Late until **9 pm the following day**; the **maximum score is 50%**. → See Week 1 submission details (p. 24)
W2	— Video 1: real hardware — Video 2: simulation without disturbance — Video 3: simulation with disturbance		Due by **9 pm** on the day of the session. Late until **9 pm the following day**; the **maximum score is 50%**. → See Week 2 submission details (p. 43)
W3	— Video 1: maximum disturbance/load on real hardware — Video 2: maximum disturbance in simulation — One-page PDF summary		Due by **9 pm** on the day of the session. Late until **9 pm the following day**; the **maximum score is 50%**. → See Week 3 submission details (p. 45)
W4	Final presentation & Q&A	**40%**	Live 10-minute group presentation; no written submission required. → See Week 4 details (p. 47)

DEADLINE

21:00

on the day of the class

Late submissions accepted until **21:00 the following day** . Late score capped at 50%.

Presentation in English

Week 1 Submission

Video (max 200 MB):

show the wheeled inverted pendulum balancing upright for at least 5 seconds.

Format	MP4 · smartphone recording recommended
Submit to	forms.gle/5huiVLJAj5Bh16Dj7 Week1_GroupX.zip

If the robot cannot yet stand, record its current state and clearly show which parts are already working.

Each group only needs one submission.

DEADLINE

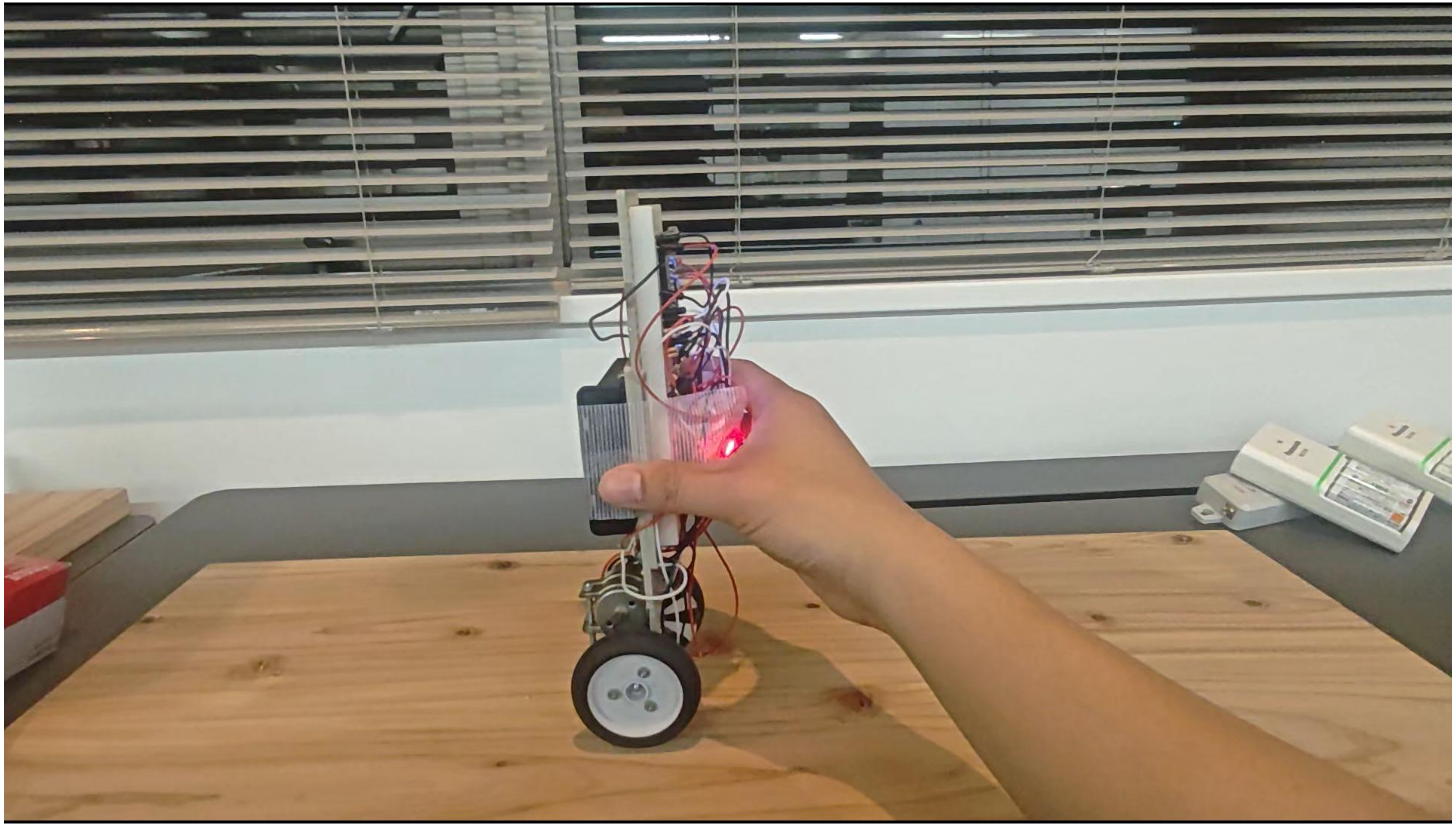
21:00

on the day of the class

Today — Monday, April 27

Late submissions accepted until **21:00 the following day** . Late score capped at 50%.

Week 1 Submission



Reminders

- If any parts are missing from the kit, please contact the TA.
- Some components in the kit may be damaged, so please test them by yourself.
- After flashing the .bin file, please restart the MCU.
- Remember to charge the battery.